

Development and Implementation of a Portable Real-Time fMRI Neurofeedback Protocol

Meghan Walsh¹, Sofia Heras¹, Julianna Wall¹, Simon Warfield¹, Alexander Li Cohen¹

¹Boston Children's Hospital, Harvard Medical School, Boston, MA

fMRI neurofeedback (fMRI-NF) offers direct evidence of neuromodulation efficacy but is hindered by complex technical barriers. We developed a user-friendly containerized real-time fMRI-NF protocol which facilitates the acquisition, processing, localization, and feedback display with low CPU/memory requirements. Our system utilizes series of containers controlled by a simple Command Line Interface, allowing real-time experimenter control with no manual configuration which can: (1) Register Atlas Region of Interest (ROI) masks to subject-space in real-time using either EasyReg or FNIRT, and (2) Read DICOM data transmitted via OS-agnostic Samba/SMB sharing from the MRI console, calculate neurofeedback metrics, and project results to the participant's screen, e.g., a rocket moving toward a portal as ROI activation increases. Graphical readout provides immediate feedback to experimenters. This portable, reproducible protocol supports fMRI-NF with commonly used Siemens (XA30) MRI scanners.